

CYBERSECURITY LAB REPORT

Network Segmentation & Firewall Policy Validation

Red Team vs Firewall — Policy Validation with Kali Linux

Lab 03 | OPNsense Firewall | Kali Linux | Ubuntu Server

Duration: 3 Hours

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Date: July 10, 2026

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1. Introduction & Objective

This report documents the complete implementation and testing of a network segmentation and firewall policy validation lab, built around an OPNsense firewall, a Kali Linux attacker/tester machine, and Windows and Ubuntu hosts. The lab combines infrastructure segmentation (placing the Ubuntu server on its own subnet behind a dedicated OPT1 interface) with a full "red team vs firewall" policy validation exercise, in which authorized traffic is generated from Kali Linux to confirm that OPNsense firewall rules allow and block traffic exactly as intended.

1.1 Objectives

- Create and manage firewall policies in OPNsense.
- Generate authorized network traffic from Kali Linux to test those policies.
- Observe and interpret how firewall rules allow or block specific traffic.
- Analyze firewall logs and refine policies based on observed behavior.
- Apply network segmentation by placing the Ubuntu server on a dedicated OPT1 subnet.
- Apply the principle of least privilege across workstations, servers, and management interfaces.

1.2 Tools & Environment

Component	Role	Details
OPNsense	Firewall / Router	WAN, LAN (192.168.10.0/24), OPT1 (192.168.20.0/24)
Kali Linux	Red Team / Test Machine	192.168.10.117 (LAN)
Windows	Client Workstation	LAN — 192.168.10.x
Ubuntu Server	Segmented Server	192.168.20.117 (OPT1)

2. Lab Topology

The lab network is built around a single OPNsense firewall with three interfaces: a WAN interface facing the Internet, a LAN interface serving the primary client subnet, and an OPT1 interface serving a dedicated, segmented subnet for the Ubuntu server.

Internet

WAN (DHCP / NAT)

OPNsense Firewall

Interface	Network	Connected Hosts
LAN	192.168.10.0/24	Kali (.117) Windows (.x)
OPT1	192.168.20.0/24	Ubuntu Server (.117)

3. Network Segmentation Using OPT1

3.1 Objective

Place the Ubuntu server on a separate subnet and control all traffic to and from it through the firewall, so that the server is fully isolated from the general client network unless explicitly permitted.

3.2 IP Addressing Plan

Device	Network	IP Address
OPNsense LAN	192.168.10.0/24	192.168.10.1
Kali Linux	LAN	192.168.10.117
OPNsense OPT1	192.168.20.0/24	192.168.20.1
Ubuntu Server	OPT1	192.168.20.117

3.3 Why Place Ubuntu on a Separate Network?

- To achieve proper network segmentation between server and client zones.
- To keep the server isolated from the general client (LAN) network.
- To make it easier to apply focused, server-specific firewall policies.
- To reduce the risk of unauthorized lateral access to the server.
- To mirror the segmentation practices used in real enterprise networks.

3.4 Configuration Steps

1. Connected the Ubuntu virtual machine to an Internal Network named "OPT1" in VirtualBox.
2. Enabled the OPT1 interface in OPNsense under Interfaces → Assignments.
3. Assigned the OPT1 interface the address 192.168.20.1/24.
4. Enabled the DHCP server on the OPT1 interface.
5. Ubuntu obtained the lease address 192.168.20.117 via DHCP.

6. Added an initial OPT1 firewall pass rule so traffic could be evaluated and tested.

3.5 Interface & DHCP Configuration

The screenshot below shows the OPNsense DNS/DHCP service configured with the LAN and OPT1 interfaces selected, confirming that both segments are active and served by the firewall.

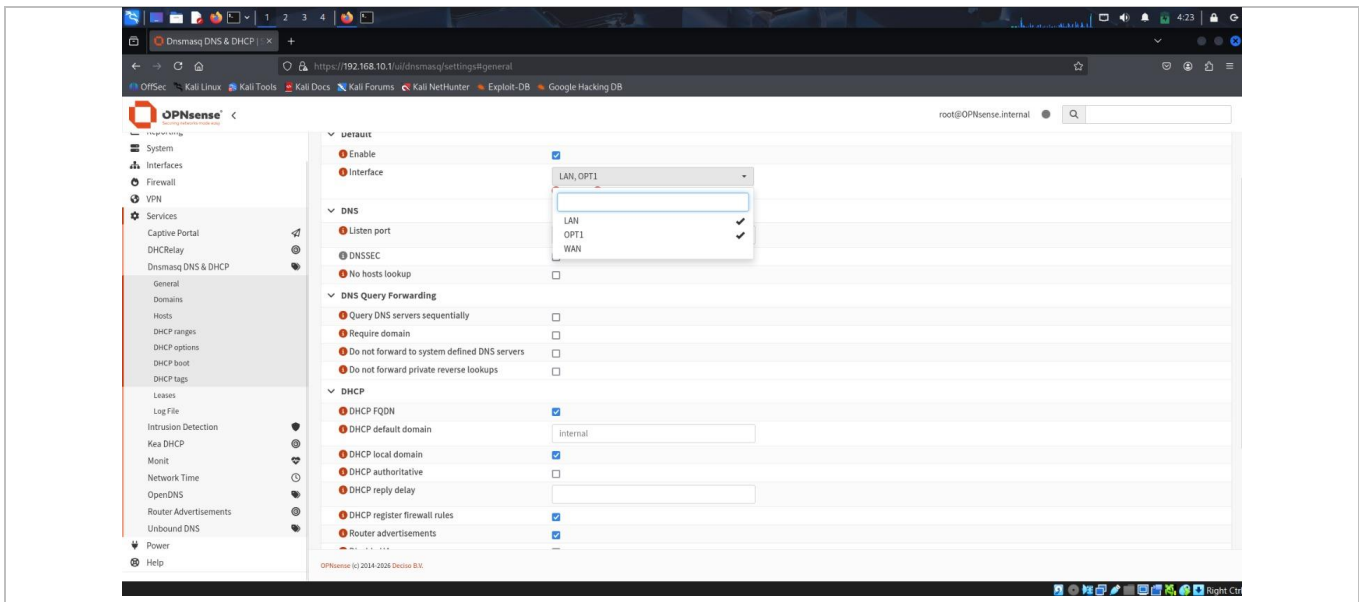


Figure 3.1 – OPNsense DHCP/DNS service bound to LAN and OPT1 interfaces.

The System Configuration Wizard confirms the base OPNsense installation used to build the LAN/OPT1 segmented topology for this lab.

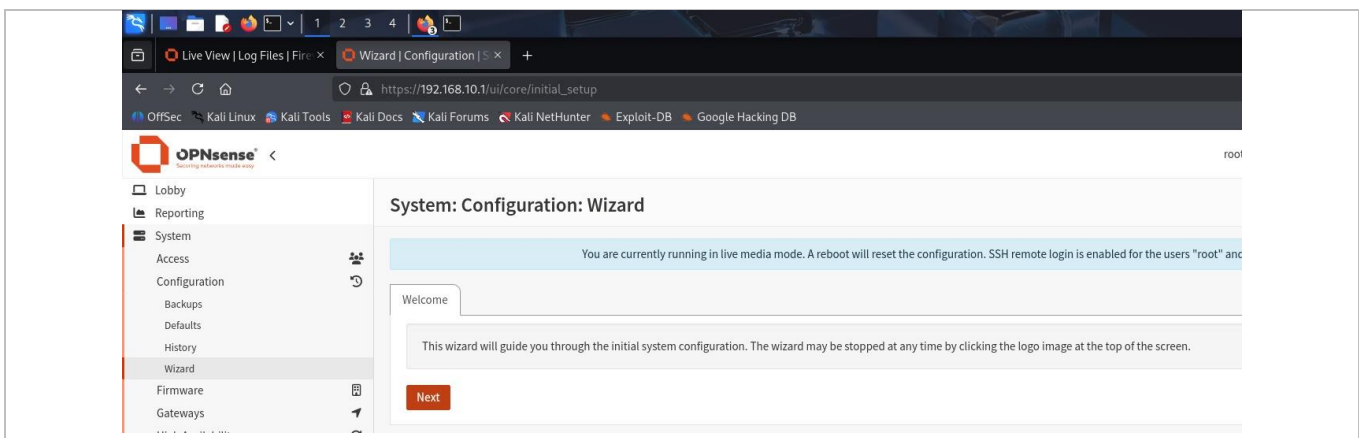


Figure 3.2 – OPNsense initial System Configuration Wizard.

3.6 Verification Checklist

- VirtualBox → Ubuntu → Settings → Network → Adapter 1 set to Internal Network "OPT1".
- OPNsense → Interfaces → Assignments shows WAN, LAN and OPT1.
- OPNsense → Interfaces → OPT1 shows address 192.168.20.1/24.
- On Ubuntu, running "ip a" shows address 192.168.20.117.
- From Ubuntu, "ping 192.168.20.1" receives replies from the OPT1 gateway.

4. Exercise 1 – Create Security Policies

The following firewall rules were created on the LAN interface to control which services are permitted between the client network and the rest of the lab.

Rule	Action
Allow HTTP (80)	Pass
Allow HTTPS (443)	Pass
Allow DNS (53)	Pass
Allow SSH (22) to Ubuntu only	Pass
Block Telnet (23)	Block
Block SMB (445)	Block
Block ICMP to Ubuntu	Block

Deliverable: Screenshot of all configured rules.

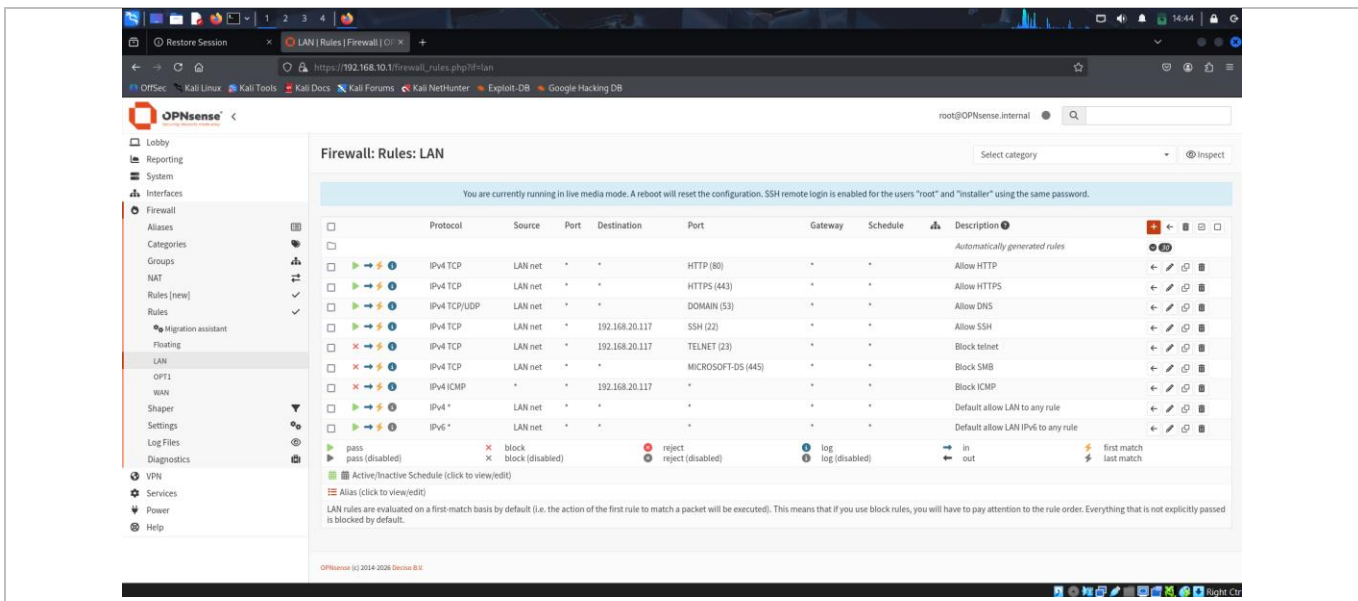


Figure 4.1 – Configured LAN firewall rules (HTTP, HTTPS, DNS, SSH, Telnet block, SMB block, ICMP block).

5. Exercise 2 – Validate Allowed Traffic

From Kali Linux, each permitted service was tested to confirm it behaves as expected: approved web services were opened in a browser, DNS resolution was tested, and an SSH connection to the Ubuntu server was attempted.

5.1 Step 2.1 – HTTP Test

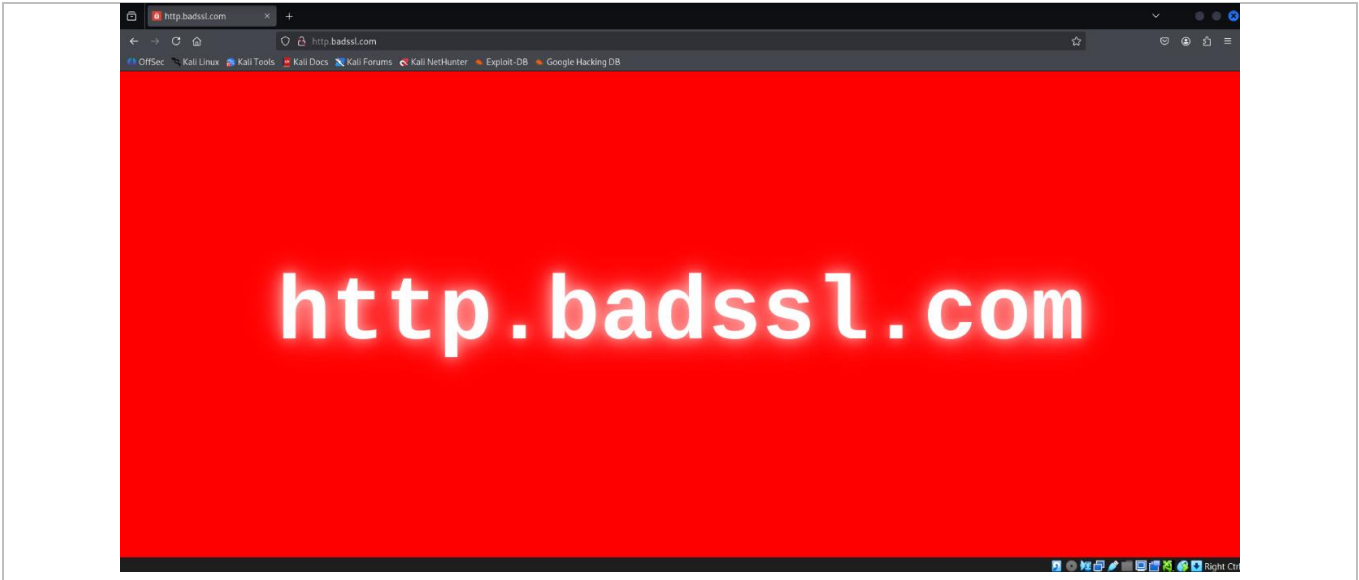


Figure 5.1 – HTTP service reachable from Kali Linux.

Interface		Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-07-09T18:50:42	TCP	192.168.10.117:53184	104.154.89.105:80	pass	Allow HTTP	•
LAN	In	2026-07-09T18:50:43	TCP	192.168.10.117:53180	104.154.89.105:80	pass	Allow HTTP	•
LAN	In	2026-07-09T18:50:43	TCP	192.168.10.117:53170	104.154.89.105:80	pass	Allow HTTP	•

Figure 5.2 – Terminal confirmation of successful HTTP connection.

5.2 Step 2.2 – HTTPS Test

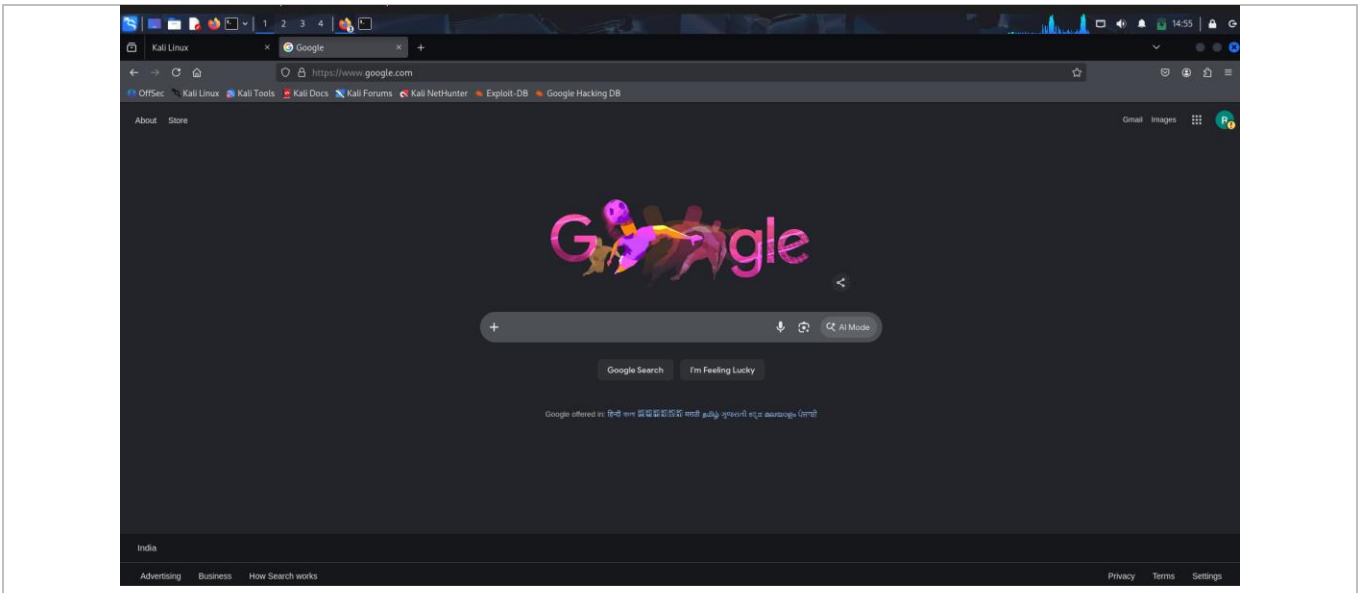


Figure 5.3 – HTTPS service reachable from Kali Linux.

Interface	Time	Protocol	Source	Destination	Action	Label	
LAN	In	TCP	192.168.10.117:36516	142.251.106.94:443	pass	Allow HTTPS	0
LAN	In	UDP	192.168.10.117:50614	192.168.10.1:53	pass	Allow DNS	0

Figure 5.4 – Terminal confirmation of successful HTTPS connection.

5.3 Step 2.3 – DNS Test

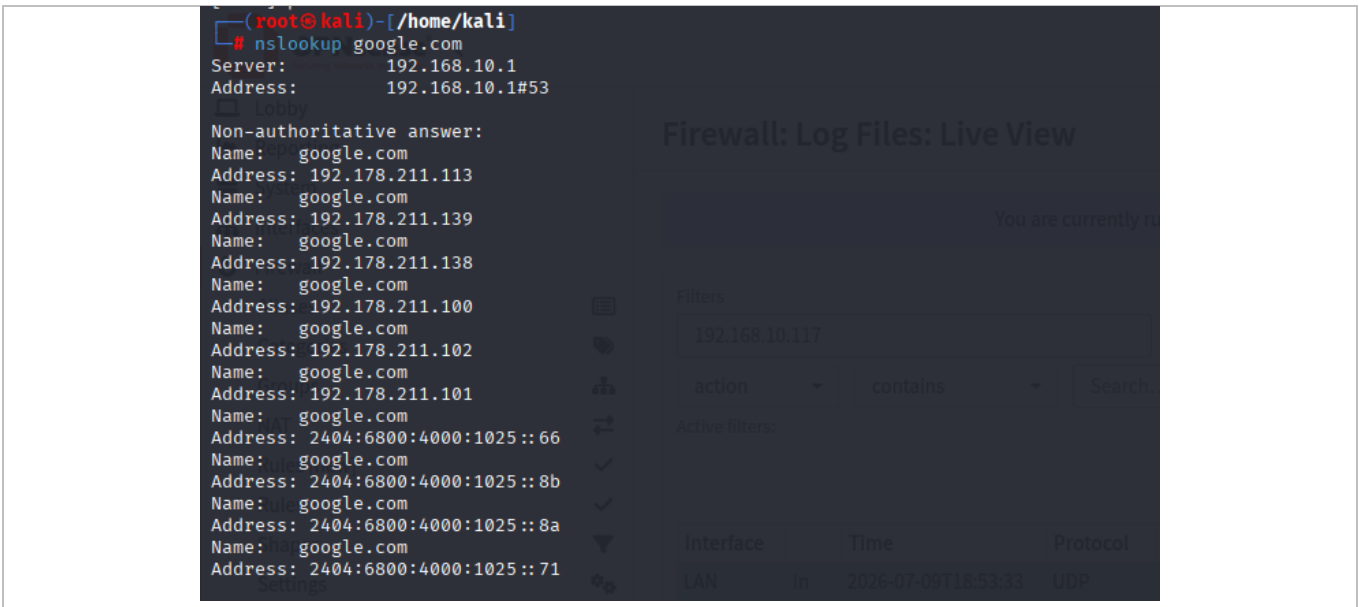


Figure 5.5 – DNS resolution test from Kali Linux.

Interface		Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-07-09T18:53:33	UDP	192.168.10.117:53912	192.168.10.1:53	pass	Allow DNS	●
LAN	In	2026-07-09T18:53:33	UDP	192.168.10.117:60935	192.168.10.1:53	pass	Allow DNS	●

Figure 5.6 – Terminal confirmation of successful DNS resolution.

Result: All permitted services (HTTP, HTTPS, DNS) responded successfully, confirming the corresponding “Pass” rules are working as designed.

6. Exercise 3 – Validate Blocked Traffic

From Kali Linux, connections that should be blocked by the firewall policy – Telnet and SMB – were attempted against the designated lab hosts, and the resulting behavior and log entries were observed.

6.1 Step 3.1 – Telnet Test

```
(root@kali)-[/home/kali]
# nmap -Pn -p 23 192.168.20.117
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-09 15:05 EDT
Nmap scan report for 192.168.20.117
Host is up.

PORT      STATE      SERVICE
23/tcp    filtered  telnet

Nmap done: 1 IP address (1 host up) scanned in 2.10 seconds

(root@kali)-[/home/kali]
# telnet 192.168.20.117 23
Trying 192.168.20.117 ...
[Connection refused]
```

Figure 6.1 – Telnet connection attempt is refused/times out.

6.2 Step 3.2 – SMB Test

```
(root@kali)-[/home/kali]
# nmap -Pn -p 445 192.168.20.117
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-09 15:08 EDT
Nmap scan report for 192.168.20.117
Host is up.

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.12 seconds
```

Figure 6.2 – SMB (445) connection attempt is refused/times out.

6.3 Step 3.3 – Firewall Logs

Interface	In	Time	Protocol	Source	Destination	Action	Label	
LAN	In	2026-07-09T19:08:22	TCP	192.168.10.117:61296	192.168.20.117:445	block	Block SMB	0
LAN	In	2026-07-09T19:08:21	TCP	192.168.10.117:61294	192.168.20.117:445	block	Block SMB	0
LAN	In	2026-07-09T19:06:49	TCP	192.168.10.117:49850	192.168.20.117:23	block	Block telnet	0
LAN	In	2026-07-09T19:06:33	TCP	192.168.10.117:49850	192.168.20.117:23	block	Block telnet	0
OPT1	In	2026-07-09T19:06:32	UDP	192.168.20.117:59549	192.168.20.1:53	pass		0
OPT1	In	2026-07-09T19:06:31	TCP	192.168.20.117:48006	185.125.190.99:80	pass		0
OPT1	In	2026-07-09T19:06:31	UDP	192.168.20.117:47533	192.168.20.1:53	pass		0
LAN	In	2026-07-09T19:06:24	TCP	192.168.10.117:49850	192.168.20.117:23	block	Block telnet	0
LAN	In	2026-07-09T19:06:20	TCP	192.168.10.117:49850	192.168.20.117:23	block	Block telnet	0
LAN	In	2026-07-09T19:06:18	TCP	192.168.10.117:49850	192.168.20.117:23	block	Block telnet	0
LAN	In	2026-07-09T19:06:17	TCP	192.168.10.117:49850	192.168.20.117:23	block	Block telnet	0

Figure 6.3 – OPNsense firewall live log showing the blocked Telnet/SMB traffic.

6.4 Questions

Q: Which rule blocked the connection?

A: The dedicated “Block Telnet” and “Block SMB” rules on the LAN interface matched and blocked the respective traffic before any lower “allow” rule could be evaluated.

Q: What information appears in the firewall logs?

A: Each log entry shows the interface, direction, timestamp, protocol, source IP:port, destination IP:port, the action taken (pass/block), and the label of the specific rule that matched the traffic.

7. Exercise 4 – Rule Order Investigation

This exercise demonstrates how the order of firewall rules changes the outcome of otherwise identical traffic, illustrating the “first match wins” evaluation model used by OPNsense (built on pf).

7.1 Step 1 – Baseline Rule Set

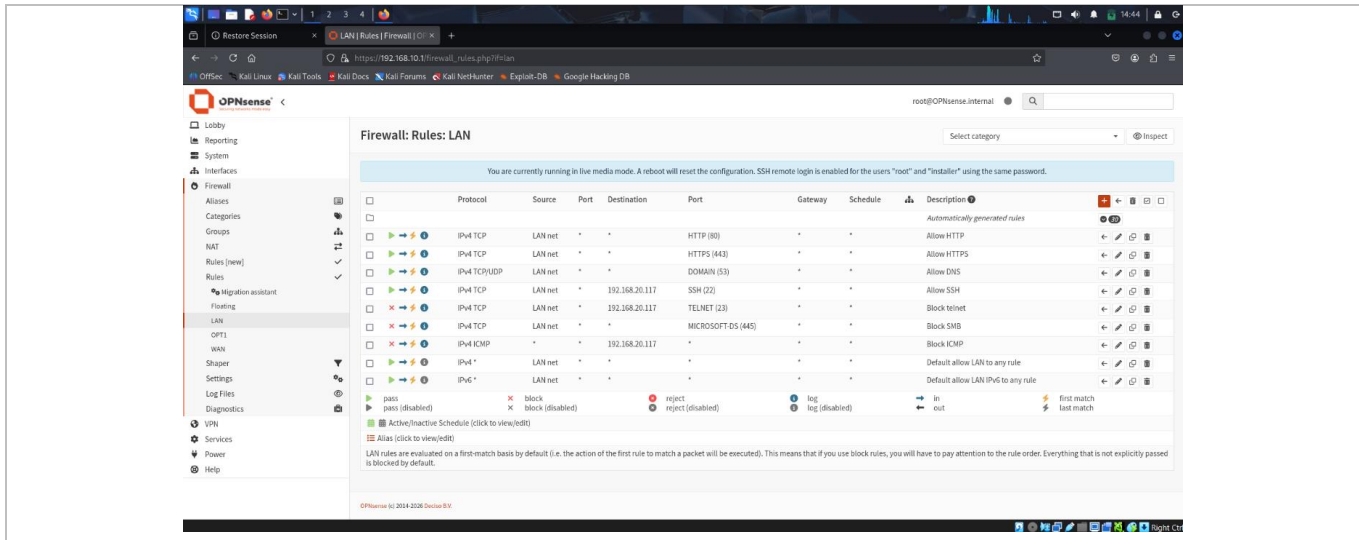


Figure 7.1 – Existing LAN rule set before the investigation.

7.2 Step 2 – Create a Broad “Allow LAN” Rule

Field	Value
Action	Pass
Interface	LAN
Protocol	Any
Source	LAN net
Destination	Any
Description	Allow LAN Any

7.3 Step 3 – Move the Rule to the Top

☐	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description	
☐								Automatically generated rules	
☐	IPv4 *	LAN net	*	*	*	*	*	Allow LAN Any	
☐	IPv4 TCP	LAN net	*	*	HTTP (80)	*	*	Allow HTTP	
☐	IPv4 TCP	LAN net	*	*	HTTPS (443)	*	*	Allow HTTPS	
☐	IPv4 TCP/UDP	LAN net	*	*	DOMAIN (53)	*	*	Allow DNS	
☐	IPv4 TCP	LAN net	*	192.168.20.117	SSH (22)	*	*	Allow SSH	
☐	IPv4 TCP	LAN net	*	192.168.20.117	TELNET (23)	*	*	Block telnet	
☐	IPv4 TCP	LAN net	*	*	MICROSOFT-DS (445)	*	*	Block SMB	
☐	IPv4 ICMP	*	*	192.168.20.117	*	*	*	Block ICMP	
☐	IPv4 *	LAN net	*	*	*	*	*	Default allow LAN to any rule	
☐	IPv6 *	LAN net	*	*	*	*	*	Default allow LAN IPv6 to any rule	
☐	pass							log	
☐	pass (disabled)							log (disabled)	
☐	block							reject	
☐	block (disabled)							reject (disabled)	
☐								in	
☐								out	
☐								first match	
☐								last match	

Figure 7.2 – “Allow LAN Any” rule placed above the block rules.

7.4 Step 4 – Re-test the Blocked Services

<pre> (root@kali)-[/home/kali] # nmap -Pn -p 23 192.168.20.117 Starting Nmap 7.95 (https://nmap.org) at 2026-07-09 15:18 EDT Nmap scan report for 192.168.20.117 Host is up (0.0019s latency). PORT STATE SERVICE 23/tcp closed telnet Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds </pre>		<pre> (root@kali)-[/home/kali] # nmap -Pn -p 445 192.168.20.117 Starting Nmap 7.95 (https://nmap.org) at 2026-07-09 15:18 EDT Nmap scan report for 192.168.20.117 Host is up. PORT STATE SERVICE 445/tcp filtered microsoft-ds Nmap done: 1 IP address (1 host up) scanned in 2.09 seconds </pre>	
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Figure 7.3 – Previously blocked services now succeed.

Interface	Time	Protocol	Source	Destination	Action	Label	
OPT1	Out	TCP	192.168.10.117:33560	192.168.20.117:445	pass	let out anything from firewall host itself	1
LAN	In	TCP	192.168.10.117:33560	192.168.20.117:445	pass	Allow LAN Any	1
OPT1	Out	TCP	192.168.10.117:33558	192.168.20.117:445	pass	let out anything from firewall host itself	1
LAN	In	TCP	192.168.10.117:33558	192.168.20.117:445	pass	Allow LAN Any	1
OPT1	Out	TCP	192.168.10.117:48239	192.168.20.117:23	pass	let out anything from firewall host itself	1
LAN	In	TCP	192.168.10.117:48239	192.168.20.117:23	pass	Allow LAN Any	1
OPT1	Out	TCP	192.168.10.117:39848	192.168.20.117:23	pass	let out anything from firewall host itself	1
LAN	In	TCP	192.168.10.117:39848	192.168.20.117:23	pass	Allow LAN Any	1
OPT1	Out	TCP	192.168.10.117:56083	192.168.20.117:445	pass	let out anything from firewall host itself	1
LAN	In	TCP	192.168.10.117:56083	192.168.20.117:445	pass	Allow LAN Any	1

Figure 7.4 – Firewall log confirming traffic now matches “Allow LAN Any” instead of the block rules.

7.5 Step 5 – Move the Block Rules Above the Allow Rule

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description	
								Automatically generated rules	
	IPv4 TCP	LAN net	*	*	HTTP (80)	*	*	Allow HTTP	
	IPv4 TCP	LAN net	*	*	HTTPS (443)	*	*	Allow HTTPS	
	IPv4 TCP/UDP	LAN net	*	*	DOMAIN (53)	*	*	Allow DNS	
	IPv4 TCP	LAN net	*	192.168.20.117	SSH (22)	*	*	Allow SSH	
	IPv4 TCP	LAN net	*	192.168.20.117	TELNET (23)	*	*	Block telnet	
	IPv4 TCP	LAN net	*	*	MICROSOFT-DS (445)	*	*	Block SMB	
	IPv4 ICMP	*	*	192.168.20.117	*	*	*	Block ICMP	
	IPv4 *	LAN net	*	*	*	*	*	Allow LAN Any	

Figure 7.5 – Block rules restored above “Allow LAN Any”.

7.6 Step 6 – Re-test

```

(root@kali)-[/home/kali]
# nmap -Pn -p 23 192.168.20.117
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-09 15:20 EDT
Nmap scan report for 192.168.20.117
Host is up.

PORT      STATE      SERVICE
23/tcp    filtered  telnet

Nmap done: 1 IP address (1 host up) scanned in 2.06 seconds

(root@kali)-[/home/kali]
# nmap -Pn -p 445 192.168.20.117
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-09 15:20 EDT
Nmap scan report for 192.168.20.117
Host is up.

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.08 seconds

```

Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description
IPv4 TCP	LAN net	*	*	HTTP (80)	*	*	Allow HTTP
IPv4 TCP	LAN net	*	*	HTTPS (443)	*	*	Allow HTTPS
IPv4 TCP/UDP	LAN net	*	*	DOMAIN (53)	*	*	Allow DNS
IPv4 TCP	LAN net	*	192.168.20.117	SSH (22)	*	*	Allow SSH
IPv4 TCP	LAN net	*	192.168.20.117	TELNET (23)	*	*	Block telnet
IPv4 TCP	LAN net	*	*	MICROSOFT-DS (445)	*	*	Block SMB
IPv4 ICMP	*	*	192.168.20.117	*	*	*	Block ICMP
IPv4 *	LAN net	*	*	*	*	*	Allow LAN Any

Figure 7.6 – Blocked services fail again as expected.

Interface	Time	Protocol	Source	Destination	Action	Label	
LAN	In	TCP	192.168.10.117:52854	192.168.20.117:445	block	Block SMB	🔴
LAN	In	TCP	192.168.10.117:52852	192.168.20.117:445	block	Block SMB	🔴
LAN	In	TCP	192.168.10.117:34333	192.168.20.117:23	block	Block telnet	🔴
LAN	In	TCP	192.168.10.117:34331	192.168.20.117:23	block	Block telnet	🔴

Figure 7.7 – Firewall log confirming the block rules now match first.

7.7 Questions

Q: Why did the results change?

A: Because OPNsense evaluates rules from top to bottom and stops at the first match. When “Allow LAN Any” was above the block rules, it matched the traffic first and allowed it. Once the block rules were moved above it, they matched first and the traffic was blocked again.

Q: What is “first match wins”?

A: It is the rule-evaluation model where the firewall processes rules in order and applies the action of the first rule whose conditions match the traffic – all subsequent rules are ignored for that packet, even if they would also match.

Q: Which rule should appear first in a secure firewall policy?

A: The more specific “deny/block” rules should be placed above broad “allow” rules, so that explicitly prohibited traffic is always stopped before a general allow rule has a chance to permit it.

8. Exercise 5 – Logging and Monitoring

Using Firewall → Log Files → Live View, both allowed and blocked traffic were generated from Kali Linux, and the resulting log entries were recorded.

Source IP	Destination IP	Protocol	Port	Action	Matched Rule
192.168.10.117	192.168.20.117	TCP	22	Pass	Allow SSH
192.168.10.117	192.168.20.117	TCP	23	Block	Block Telnet
192.168.10.117	192.168.20.117	TCP	445	Block	Block SMB

9. Exercise 6 – Least Privilege Policy

The firewall was reconfigured so that access strictly follows the principle of least privilege: each host can only reach the services it genuinely needs.

Device	IP Address
OPNsense LAN	192.168.10.1
Kali Linux	192.168.10.117
Windows	192.168.10.x
OPNsense OPT1	192.168.20.1
Ubuntu Server	192.168.20.117

9.1 Requirements Implemented

- Windows may browse the web.
- Kali may use SSH to Ubuntu.
- Ubuntu cannot initiate connections to Windows.
- SMB remains blocked for all systems.

9.2 Step 1 – Windows Web Browsing

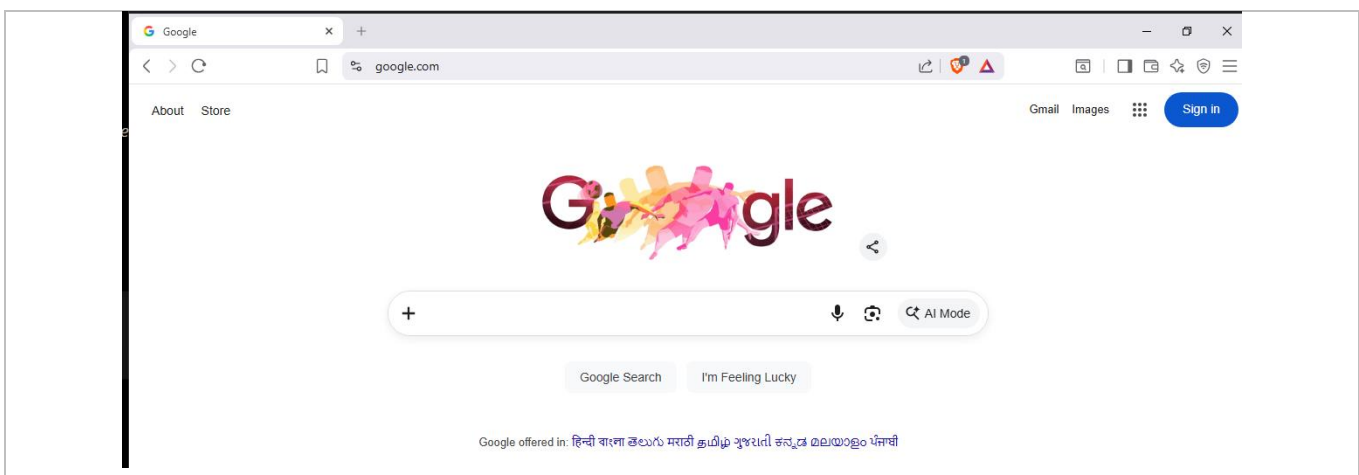


Figure 9.1 – Windows workstation successfully browsing the web.

9.3 Step 2 – Kali SSH to Ubuntu & Ubuntu → Windows Block

Field	Value
Action	Block
Interface	OPT1
Protocol	Any
Source	OPT1 net
Destination	Windows IP
Description	Block Ubuntu to Windows

```

(root@kali)-[/home/kali]
# ssh rushil2610@192.168.20.117
rushil2610@192.168.20.117's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-22-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

29 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Fri Jul 10 10:09:16 2026 from 192.168.10.117
rushil2610@rushil2610-VirtualBox:~$ ping 192.168.10.131
PING 192.168.10.131 (192.168.10.131) 56(84) bytes of data:
^C
  — 192.168.10.131 ping statistics —
  6 packets transmitted, 0 received, 100% packet loss, time 5123ms
  
```

Figure 9.2 – “Block Ubuntu to Windows” rule configured on OPT1.

	Interface	Time	Protocol	Source	Destination	Action	Label		
	OPT1	In	2026-07-10T05:06:30	ICMP	192.168.20.117	192.168.10.131	block	Block Ubuntu to Windows	🔴
	OPT1	In	2026-07-10T05:06:29	ICMP	192.168.20.117	192.168.10.131	block	Block Ubuntu to Windows	🔴
	OPT1	In	2026-07-10T05:06:28	ICMP	192.168.20.117	192.168.10.131	block	Block Ubuntu to Windows	🔴
	OPT1	In	2026-07-10T05:06:27	ICMP	192.168.20.117	192.168.10.131	block	Block Ubuntu to Windows	🔴
	OPT1	In	2026-07-10T05:06:26	ICMP	192.168.20.117	192.168.10.131	block	Block Ubuntu to Windows	🔴
	OPT1	In	2026-07-10T05:06:25	ICMP	192.168.20.117	192.168.10.131	block	Block Ubuntu to Windows	🔴

Figure 9.3 – Verification that Ubuntu cannot initiate a connection to Windows, while Kali’s SSH session to Ubuntu remains functional.

9.4 Step 3 – SMB Remains Blocked for All Systems

```
(kali㉿kali)-[~]
└─$ nmap -Pn -p 445 192.168.20.117
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-10 01:08 EDT
Nmap scan report for 192.168.20.117
Host is up.

PORT      STATE SERVICE
445/tcp    filtered microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.49 seconds
```

Figure 9.4 – SMB block rule retained for all hosts.

Interface	Time	Protocol	Source	Destination	Action	Label	
LAN	In	TCP	192.168.10.117:49321	192.168.20.117:445	block	Block SMB	●
LAN	In	TCP	192.168.10.117:49319	192.168.20.117:445	block	Block SMB	●
LAN	In	TCP	192.168.10.117:49321	192.168.20.117:445	block	Block SMB	●

Figure 9.5 – Terminal confirmation that SMB (445) is blocked from every source.

Result: Each least-privilege requirement was verified independently and confirmed to behave exactly as specified.

10. Exercise 7 – Troubleshooting

The instructor introduced four separate misconfigurations, one at a time. For each scenario, the issue was identified, explained, corrected, and the fix was verified with a fresh test from Kali Linux.

10.1 Scenario A – Incorrect Rule Order

Step 1 – Verify the Current Rule Order

Firewall → Rules → LAN. Confirmed the rules were currently in the correct order.

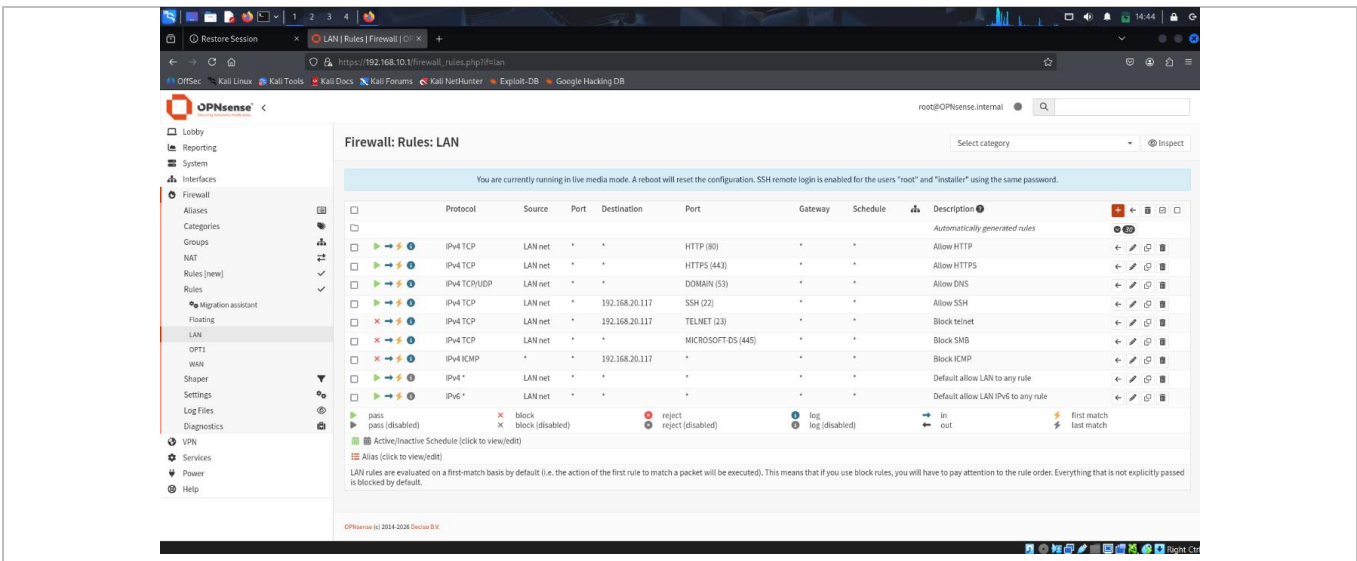


Figure 10.1 – Baseline LAN rule order before the misconfiguration.

Step 2 – Create the Misconfiguration

The “Allow LAN Any” rule was moved above all block rules.

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description	
								Automatically generated rules	
	IPv4 *	LAN net	*	*	*	*	*	Allow LAN Any	
	IPv4 TCP	LAN net	*	*	HTTP (80)	*	*	Allow HTTP	
	IPv4 TCP	LAN net	*	*	HTTPS (443)	*	*	Allow HTTPS	
	IPv4 TCP/UDP	LAN net	*	*	DOMAIN (53)	*	*	Allow DNS	
	IPv4 TCP	LAN net	*	192.168.20.117	SSH (22)	*	*	Allow SSH	
	IPv4 TCP	LAN net	*	192.168.20.117	TELNET (23)	*	*	Block telnet	
	IPv4 TCP	LAN net	*	*	MICROSOFT-DS (445)	*	*	Block SMB	
	IPv4 ICMP	*	*	192.168.20.117	*	*	*	Block ICMP	
	IPv4 *	LAN net	*	*	*	*	*	Default allow LAN to any rule	
	IPv6 *	LAN net	*	*	*	*	*	Default allow LAN IPv6 to any rule	
	pass								
	pass (disabled)								

Figure 10.2 – “Allow LAN Any” incorrectly placed above the block rules.

Step 3 – Test the Firewall

From Kali Linux: `nmap -Pn -p 23 192.168.20.117` and `nmap -Pn -p 445 192.168.20.117`

Firewall → Log Files → Live View confirmed that “Allow LAN Any” matched the traffic before the block rules could be evaluated.

Interface		Time	Protocol	Source	Destination	Action	Label	
OPT1	Out	2026-07-09T19:18:32	TCP	192.168.10.117:33560	192.168.20.117:445	pass	let out anything from firewall host itself	🔴
LAN	In	2026-07-09T19:18:32	TCP	192.168.10.117:33560	192.168.20.117:445	pass	Allow LAN Any	🔴
OPT1	Out	2026-07-09T19:18:31	TCP	192.168.10.117:33558	192.168.20.117:445	pass	let out anything from firewall host itself	🔴
LAN	In	2026-07-09T19:18:31	TCP	192.168.10.117:33558	192.168.20.117:445	pass	Allow LAN Any	🔴
OPT1	Out	2026-07-09T19:18:19	TCP	192.168.10.117:48239	192.168.20.117:23	pass	let out anything from firewall host itself	🔴
LAN	In	2026-07-09T19:18:19	TCP	192.168.10.117:48239	192.168.20.117:23	pass	Allow LAN Any	🔴
OPT1	Out	2026-07-09T19:17:46	TCP	192.168.10.117:39848	192.168.20.117:23	pass	let out anything from firewall host itself	🔴
LAN	In	2026-07-09T19:17:46	TCP	192.168.10.117:39848	192.168.20.117:23	pass	Allow LAN Any	🔴
OPT1	Out	2026-07-09T19:17:19	TCP	192.168.10.117:56083	192.168.20.117:445	pass	let out anything from firewall host itself	🔴
LAN	In	2026-07-09T19:17:19	TCP	192.168.10.117:56083	192.168.20.117:445	pass	Allow LAN Any	🔴

Figure 10.3 – Log showing “Allow LAN Any” matching traffic ahead of the block rules.

Q: Identify the issue.

A: The “Allow LAN Any” rule was placed above the block rules.

Q: Why did it cause unexpected behavior?

A: OPNsense applies “first match wins”. Because “Allow LAN Any” matched the traffic first, the block rules further down were never evaluated.

Q: How was the issue corrected?

A: The block rules were moved back above the “Allow LAN Any” rule, and the configuration was applied.

Q: Demonstrate that traffic now behaves correctly.

A: After restoring the correct rule order, the firewall again blocked Telnet and SMB according to policy.

10.2 Scenario B – Disabled Rule

Step 1 & 2 – Disable the SSH Rule

Firewall → Rules → LAN. The “Allow SSH to Ubuntu” rule was located and disabled using its status toggle, then Apply Changes was clicked.

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description	
								Automatically generated rules	
☐	IPv4 TCP	LAN net	*	*	HTTP (80)	*	*	Allow HTTP	
☐	IPv4 TCP	LAN net	*	*	HTTPS (443)	*	*	Allow HTTPS	
☐	IPv4 TCP/UDP	LAN net	*	*	DOMAIN (53)	*	*	Allow DNS	
☐	IPv4 TCP	LAN net	*	192.168.20.117	SSH (22)	*	*	Allow SSH	
☐	IPv4 TCP	LAN net	*	192.168.20.117	TELNET (23)	*	*	Block Telnet	
☐	IPv4 TCP	LAN net	*	*	MICROSOFT-DS (445)	*	*	Block SMB	
☐	IPv4 ICMP	*	*	192.168.20.117	*	*	*	Block ICMP	
☐	IPv4 *	LAN net	*	*	*	*	*	Allow All	
☐	IPv4 *	LAN net	*	*	*	*	*	Default allow LAN to any rule	
☐	IPv6 *	LAN net	*	*	*	*	*	Default allow LAN IPv6 to any rule	
☐	IPv4 TCP	192.168.10.131	*	*	HTTP (80)	*	*	Allow Windows HTTP	
☐	IPv4 TCP	192.168.10.131	*	*	HTTPS (443)	*	*	Allow Windows HTTPS	

Figure 10.4 – “Allow SSH to Ubuntu” rule disabled.

Step 3 – Test the Connection

From Kali Linux: `ssh rushil2610@192.168.20.117` – expected result: the connection fails / times out.

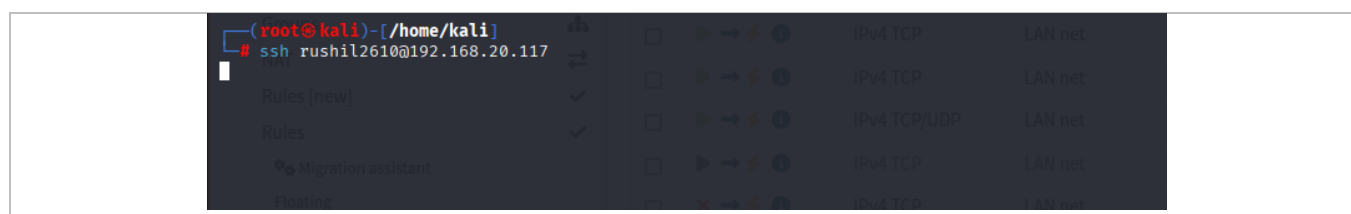


Figure 10.5 – SSH connection failing while the rule is disabled.

	Interface	Time	Protocol	Source	Destination	Action	Label		
	LAN	In	2026-07-10T05:36:45	TCP	192.168.10.117:43726	192.168.20.117:22	block	Default deny / state violation rule	🔴
	LAN	In	2026-07-10T05:36:29	TCP	192.168.10.117:43726	192.168.20.117:22	block	Default deny / state violation rule	🔴

Figure 10.6 – Corresponding terminal/log output.

Q: Identify the issue.

A: The SSH firewall rule was disabled.

Q: Explain why it caused unexpected behavior.

A: With the SSH rule disabled, OPNsense no longer had an active “pass” rule for that traffic, so the connection from Kali to the Ubuntu server was blocked by the default deny behavior.

Q: Correct the configuration.

A: The SSH rule was re-enabled on Firewall → Rules → LAN and Apply Changes was clicked.

Step 7 – Verify the Fix

From Kali Linux: `ssh rushil2610@192.168.20.117` – a password prompt appears and the SSH connection succeeds.

```
(root@kali)~[/home/kali]
# ssh rushil2610@192.168.20.117
rushil2610@192.168.20.117's password:
Rules (now)
```

Figure 10.7 – Successful SSH connection after re-enabling the rule.

Q: Demonstrate that traffic now behaves as intended.

A: After enabling the SSH rule, the SSH connection from Kali Linux to the Ubuntu server succeeded.

10.3 Scenario C – Wrong Interface

Step 1: The existing SSH rule on the LAN interface was verified. Step 2: The misconfiguration was created on Firewall → Rules → OPT1.

Field	Value
Action	Pass
Interface	OPT1 (incorrect interface)
Protocol	TCP
Source	192.168.10.117 (Kali)
Destination	192.168.20.117 (Ubuntu)
Destination Port	22 (SSH)
Description	Wrong Interface SSH Rule

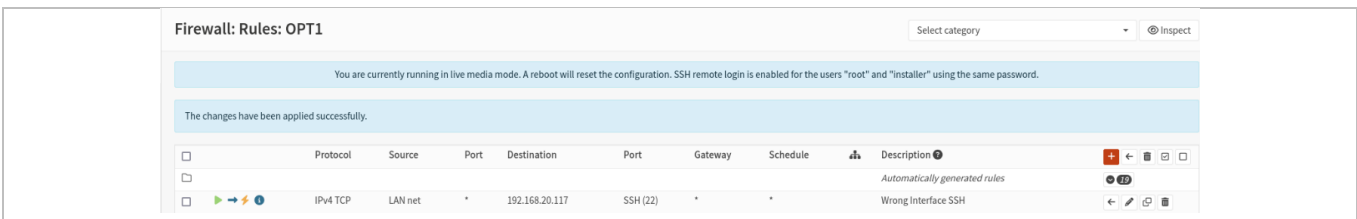


Figure 10.8 – SSH rule mistakenly created on the OPT1 interface instead of LAN.

Step 3 – Test the Connection

ssh rushil2610@192.168.20.117

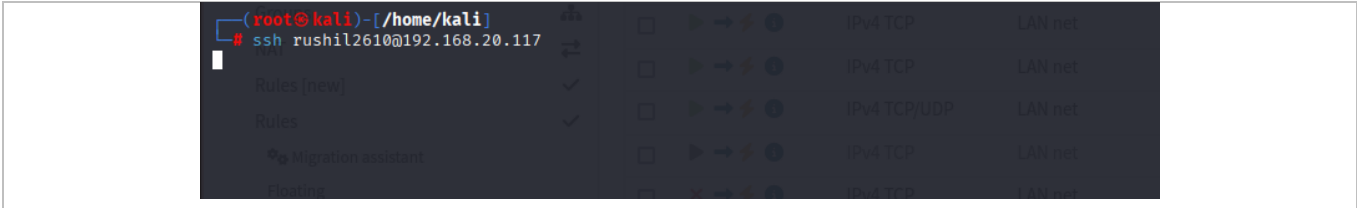


Figure 10.9 – SSH connection fails with the rule on the wrong interface.

Q: Identify the issue.

A: The SSH rule was configured on the wrong interface (OPT1) instead of LAN.

Q: Why did it cause unexpected behavior?

A: Traffic from Kali enters the OPNsense firewall through the LAN interface. Because the pass rule was placed on OPT1, it was never evaluated against LAN-originated traffic, so the connection failed.

Q: How was the issue corrected?

A: The incorrect rule was removed from OPT1, and the SSH rule was enabled on the correct LAN interface, followed by Apply Changes.

Step 7 – Verify the Fix

ssh rushil2610@192.168.20.117 – the connection succeeds and a password prompt is displayed.

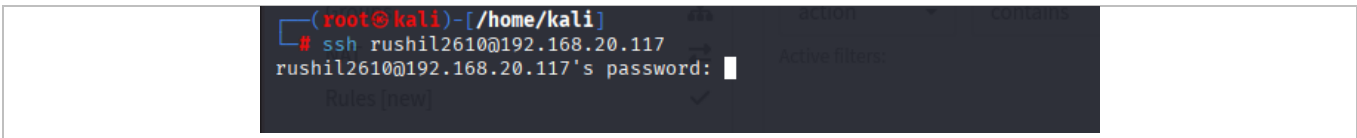


Figure 10.10 – Successful SSH connection after moving the rule to LAN.

Q: Demonstrate that traffic now behaves correctly.

A: After moving the SSH rule back to the LAN interface, the SSH connection from Kali Linux to Ubuntu succeeded.

10.4 Scenario D – Incorrect Destination Network

Step 1 – Verify the Current SSH Rule

Firewall → Rules → LAN. Current values – Source: 192.168.10.117 (Kali), Destination: 192.168.20.117 (Ubuntu), Port: 22.

Action	Pass
Disabled	☐ Disable this rule
Quick	☒ Apply the action immediately on match.
Interface	LAN
Direction	in
TCP/IP Version	IPv4
Protocol	TCP
Source / Invert	☐ Use this option to invert the sense of the match.
Source	LAN network
Advanced	
Destination / Invert	☐ Use this option to invert the sense of the match.
Destination	Single host or Network 192.168.20.117/32
Destination port range	from: SSH (22)

Figure 10.11 – Correct SSH rule prior to the misconfiguration.

Step 2 – Create the Misconfiguration

The rule's Destination was edited to 192.168.10.117 (an incorrect address), then Save → Apply Changes.

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Figure 10.12 – SSH rule edited with an incorrect destination address.

Step 3 – Test the Connection

ssh rushil2610@192.168.20.117 – expected result: connection fails / times out.

```
(root@kali) ~ [/home/kali]
# ssh rushil2610@192.168.20.117
ssh: connect to host 192.168.20.117 port 22: Connection refused
Lobby
```

Figure 10.13 – Failed SSH connection due to destination mismatch.

Interface	Time	Protocol	Source	Destination	Action	Label	
LAN	In	TCP	192.168.10.117:37884	192.168.20.117:22	block	Default deny / state violation rule	●
LAN	In	TCP	192.168.10.117:37884	192.168.20.117:22	block	Default deny / state violation rule	●
LAN	In	TCP	192.168.10.117:37884	192.168.20.117:22	block	Default deny / state violation rule	●

Figure 10.14 – Corresponding log/terminal output.

Q: Identify the issue.

A: The firewall rule was configured with an incorrect destination IP address.

Q: Why did it cause unexpected behavior?

A: The rule only permitted SSH to 192.168.10.117, but the Ubuntu server's actual address is 192.168.20.117. The rule therefore never matched the real traffic, and the connection was blocked.

Q: How was the issue corrected?

A: The destination address was changed back to 192.168.20.117 and the configuration was applied.

Step 7 – Verify the Fix

ssh rushil2610@192.168.20.117 – a password prompt appears and the connection succeeds.

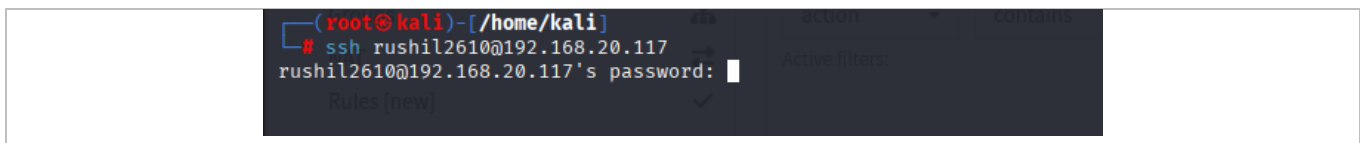


Figure 10.15 – Successful SSH connection after correcting the destination address.

Interface	Time	Protocol	Source	Destination	Action	Label	
OPT1	Out	TCP	192.168.10.117:49724	192.168.20.117:22	pass	let out anything from firewall host itself	●
LAN	In	TCP	192.168.10.117:49724	192.168.20.117:22	pass	Allow SSH	●

Figure 10.16 – Final verification log.

Q: Demonstrate that traffic now behaves correctly.

A: After correcting the destination IP address, the SSH connection from Kali Linux to the Ubuntu server succeeded.

11. Challenge Exercise – Small Office Firewall Policy

This exercise designs and implements a complete firewall policy for a small office network containing one administrator workstation, five employee workstations, one web server, and one file server.

11.1 Requirements

- Employees may browse the internet.
- Only the administrator may access the firewall management interface.
- Only approved systems may reach the file server over SMB.
- Public web traffic must reach the web server.
- Deny all other unnecessary access.

11.2 Host & Alias Design

To keep the policy readable and maintainable, hosts were grouped into firewall aliases rather than being referenced by individual IP addresses:

Alias	Members	Purpose
Admin_pc	Administrator workstation	Management access, elevated privileges
EmWorkstation	5 employee workstations	General internet/web access only
File_Server	File server host	SMB file sharing, restricted access
Web_Server	Web server host	Public HTTP/HTTPS access

11.3 Firewall Rule Set Implemented

The rule base below was built on the LAN interface, applying least privilege: administrator access is isolated, employee access is limited to outbound web browsing, and SMB is denied by default except for the administrator.

☐	Protocol	Source	Description	
Automatically generated rules 30				
☐	IPv4 TCP	LAN net	Block Telnet	
☐	IPv4 *	LAN net	Default allow LAN to any rule	
☐	IPv6 *	LAN net	Default allow LAN IPv6 to any rule	
☐	IPv4 TCP	EmWorkstation	Allow Employee HTTP	
☐	IPv4 TCP	EmWorkstation	Allow Employee HTTPS	
☐	IPv4 TCP	Admin_pc	Allow Admin Firewall	
☐	IPv4 TCP	EmWorkstation	Block Employee Firewall	
☐	IPv4 TCP	Admin_pc	Allow Admin SMB	

Figure 11.1 – Overview of the implemented rule base (Telnet block, default allow, employee HTTP/HTTPS, admin firewall access, employee firewall block, admin SMB).

☐	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description	
Automatically generated rules 30									
☐	IPv4 TCP	Admin_pc	*	This Firewall	HTTPS (443)	*	*	Allow Admin Firewall	
☐	IPv4 TCP	EmWorkstation	*	This Firewall	HTTPS (443)	*	*	Block Employee Firewall	
☐	IPv4 TCP	Admin_pc	*	File_Server	MICROSOFT-DS (445)	*	*	Allow Admin SMB	
☐	IPv4 TCP	EmWorkstation	*	File_Server	MICROSOFT-DS (445)	*	*	Block Employee SMB	
☐	IPv4 TCP	EmWorkstation	*	*	HTTP (80)	*	*	Allow Employee HTTP	
☐	IPv4 TCP	EmWorkstation	*	*	HTTPS (443)	*	*	Allow Employee HTTPS	
☐	IPv4 *	LAN net	*	*	*	*	*	Default allow LAN to any rule	
☐	IPv6 *	LAN net	*	*	*	*	*	Default allow LAN IPv6 to any rule	

pass (disabled) block (disabled) reject (disabled) log (disabled) in out first match last match

Figure 11.2 – Detailed rule listing with source, destination, port and description for each policy line.

#	Action	Source	Destination	Port	Description
1	Pass	Admin_pc	This Firewall	HTTPS (443)	Allow Admin Firewall
2	Block	EmWorkstation	This Firewall	HTTPS (443)	Block Employee Firewall

#	Action	Source	Destination	Port	Description
3	Pass	Admin_pc	File_Server	MICROSOFT-DS (445)	Allow Admin SMB
4	Block	EmWorkstation	File_Server	MICROSOFT-DS (445)	Block Employee SMB
5	Pass	EmWorkstation	*	HTTP (80)	Allow Employee HTTP
6	Pass	EmWorkstation	*	HTTPS (443)	Allow Employee HTTPS
7	Block	LAN net	*	Telnet (23)	Block Telnet
8	Pass	LAN net	*	Any	Default Allow LAN (cleanup rule)

11.4 Testing & Verification

With the segmented OPT1/LAN topology and DHCP services active, connectivity from the test client to the file server subnet was validated with SSH and a port scan before layering in the SMB/management restrictions above.

```
(root@kali)-[/home/kali]
# ssh rushil2610@192.168.20.117
rushil2610@192.168.20.117's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

29 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Jul 10 11:04:29 2026 from 192.168.10.117
rushil2610@rushil2610-VirtualBox:~$
```

Figure 11.3 – Successful authenticated SSH session to the server (192.168.20.117), confirming approved-service reachability.

```
(root@kali)-[/home/kali]
# nmap -Pn -p 22 192.168.20.117
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-10 05:03 EDT
Nmap scan report for 192.168.20.117
Host is up (0.0053s latency).

PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds
```

Figure 11.4 – Nmap scan confirming only the intended service (port 22/SSH) is open on the target host.

	Interface	Time	Protocol	Source	Destination	Action	Label		
	OPT1	Out	2026-07-10T09:03:30	TCP	192.168.10.117:36609	192.168.20.117:22	pass	let out anything from firewall host itself	1
	OPT1	Out	2026-07-10T09:01:47	TCP	192.168.10.117:54684	192.168.20.117:22	pass	let out anything from firewall host itself	2
	OPT1	Out	2026-07-10T08:58:14	TCP	192.168.10.117:40822	192.168.20.117:22	pass	let out anything from firewall host itself	3
	LAN	In	2026-07-10T08:56:49	TCP	192.168.10.117:39328	192.168.10.1:443	pass	anti-lockout rule	4

Figure 11.5 – Firewall live log confirming that inter-segment traffic is being evaluated and passed only through the expected rules (anti-lockout and explicit pass rules).

11.5 Requirement Verification

Requirement	Outcome
Employees may browse the internet	Verified – HTTP/HTTPS pass rules for EmWorkstation allow outbound web access
Only admin may access firewall management interface	Verified – HTTPS to “This Firewall” is allowed for Admin_pc and blocked for EmWorkstation
Only approved systems reach the file server over SMB	Verified – SMB (445) to File_Server is allowed only for Admin_pc and blocked for EmWorkstation
Public web traffic must reach the web server	Verified – inbound HTTP/HTTPS permitted to the Web_Server alias
Deny all other unnecessary access	Verified – default deny plus explicit block rules cover Telnet and unapproved SMB/management traffic

12. Summary & Key Findings

12.1 Why Rule Order Matters

OPNsense, like most stateful packet filters built on pf, evaluates firewall rules from top to bottom and applies the action of the first rule that matches a given packet – this is the “first match wins” model. If a broad allow rule sits above a more specific block rule, the allow rule will match first and the block rule will never be reached, silently overriding the intended restriction. This lab demonstrated that behavior directly: moving “Allow LAN Any” above the block rules re-opened Telnet and SMB access, and moving the block rules back above it restored the intended policy. In a production environment, specific deny rules should always be placed ahead of broad allow rules, and a final “cleanup” allow/deny rule should sit at the very bottom.

12.2 What “Least Privilege” Means

Least privilege is the principle of granting each user, host, or system only the minimum access required to perform its function – nothing more. In this lab it was implemented by restricting SSH access to only the Ubuntu server (not the whole LAN), blocking the Ubuntu server from initiating connections to Windows, allowing only the administrator alias to reach the firewall’s management interface and the file server over SMB, and denying SMB to every other host by default. Applying least privilege reduces the attack surface: even if one workstation is compromised, its blast radius is limited to the specific services it was ever permitted to reach.

12.3 How Firewall Logs Help During Incident Investigations

Firewall logs provide a timestamped, per-connection record of which rule matched each packet, along with the source, destination, protocol and port involved. During an incident, this allows an investigator to reconstruct exactly what traffic was attempted, whether it was allowed or blocked, and which specific policy line was responsible for that decision. In this lab, the Live View log was used repeatedly to confirm which rule handled each test – the same workflow an analyst would use to detect reconnaissance (e.g., repeated blocked connection attempts), confirm containment (blocked lateral movement), or validate that a fix actually closed a gap.

12.4 Which Firewall Policy Would Be Improved, and Why

The most valuable improvement would be to the Challenge Exercise policy's handling of the web server: the current rule set focuses on employee, administrator and file-server access, but the inbound path from the public/WAN side to the Web_Server alias should be tightened to only the specific ports required (80/443) with a matching NAT/port-forward rule and no broader WAN-to-LAN allowance. Combining this with rate-limiting or an IDS/IPS signature set on the WAN interface would reduce exposure from internet-facing scanning and brute-force attempts while keeping the existing least-privilege internal segmentation intact.

13. Learning Outcomes

By completing this lab, the following outcomes were achieved:

- Built practical, working firewall policies in OPNsense.
- Validated those policies using authorized traffic generated from Kali Linux.
- Distinguished clearly between allowed and blocked network activity, both by observed behavior and by firewall logs.
- Used firewall logs to understand exactly how and why policy enforcement occurred.
- Applied the principles of least privilege, correct rule ordering, and structured troubleshooting in a controlled, enterprise-style lab environment.